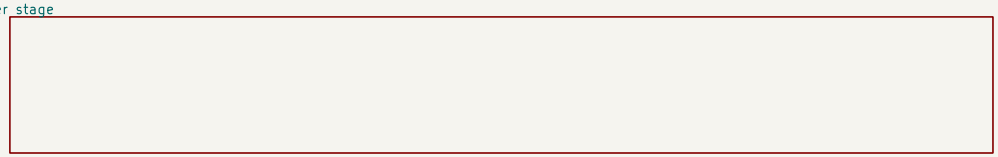
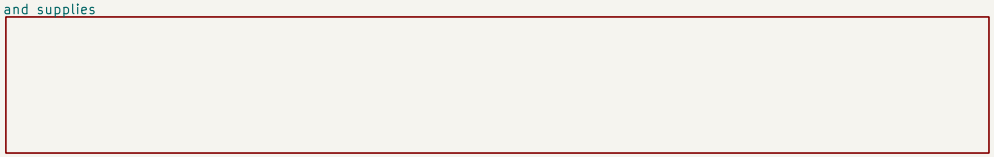
Power stage



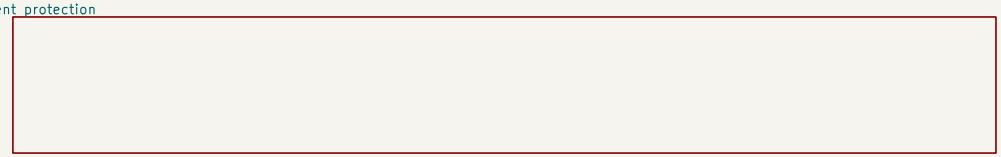
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Gate driver and supplies



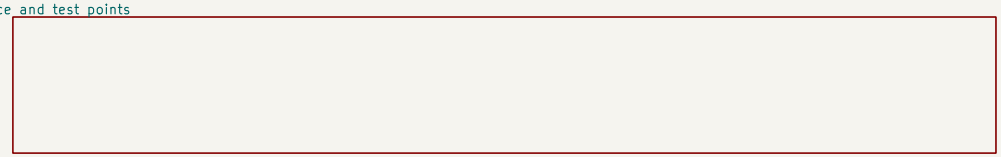
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Independent protection



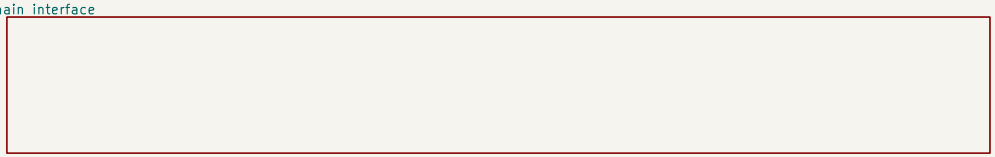
File: buck_3.kicad_sch

STM32 interface and test points



File: buck_4.kicad_sch

Power-domain interface



File: buck_5.kicad_sch

UNAL Manizales

Sheet: /
File: buck.kicad_sch

Title: Buck 12V / 6V – REV B PROTOTYPE

Size: A4	Date:	**Rev: B**
KiCad E.D.A. 10.0.7		Id: 1/6

J1

https://www.mouser.com/product/resource/TB002-500.pdf

12V IN TB002-500-02BE

Q1

https://www.infineon.com/assets/row/public/documents/24/49/infineon-irfz44n-datasheet-en.pdf

IRFZ44NPBF

D1

SS34

L1

https://www.murata.com/products/power/power-magnetics (KMP_6000B_B01)

330uH 60B334C

L2

https://www.murata.com/products/power/power-magnetics (KMP_6000B_B01)

330uH 60B334C

RS1

0.050R 1% 2W RLP25FEER050

J2

https://www.mouser.com/product/resource/TB002-500.pdf

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C1

100uF 25V

C2

100uF 25V

C3

100nF 50V

C4

10uF 50V 50SVF10M

C5

10uF 50V 50SVF10M

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UNAL Manizales – Sistemas Dinamicos y Control

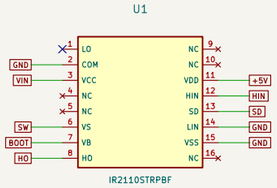
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File: buck_1.kicad_sch

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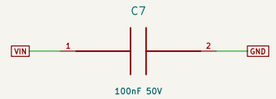
Size: A3 | Date: | Rev: B

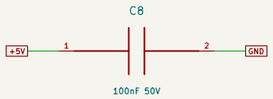
KiCad E.D.A. 10.0.7 | Id: 2/6

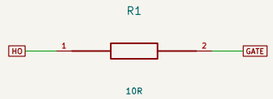


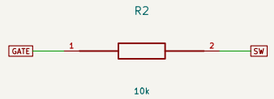


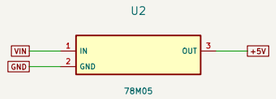


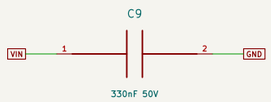




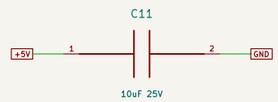


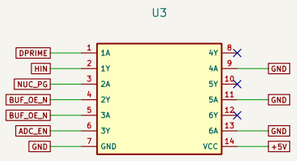


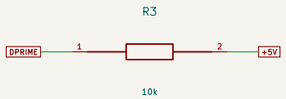


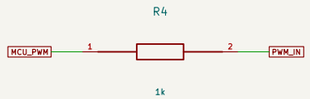


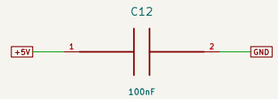










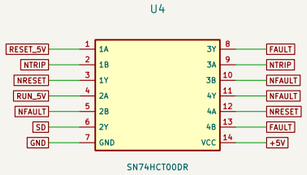


UNAL Manizales – Sistemas Dinamicos y Control

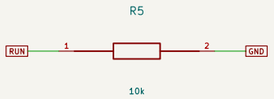
Sheet: /Gate driver and supplies/
File: buck_2.kicad_sch

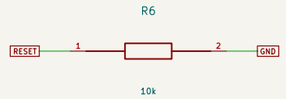
Title: Gate driver and supplies – REV B PROTOTYPE

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KiCad E.D.A. 10.0.7		Id: 3/6

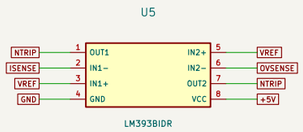




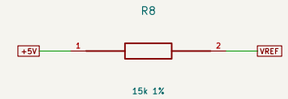


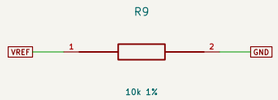


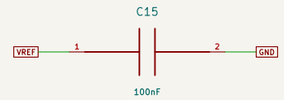




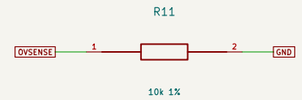


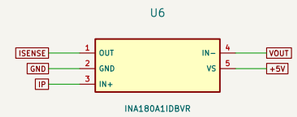




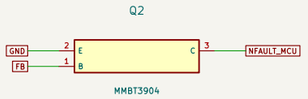


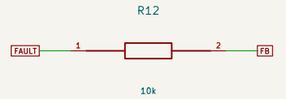


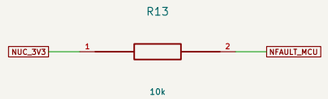












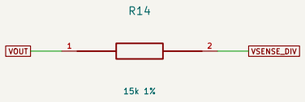
UNAL Manizales – Sistemas Dinamicos y Control

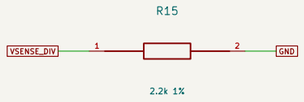
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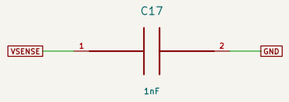
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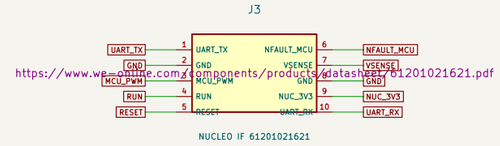
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KiCad E.D.A. 10.0.7		Id: 4/6

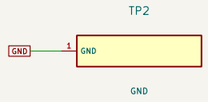


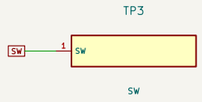


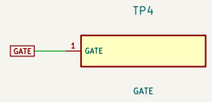


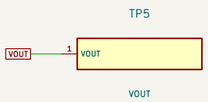


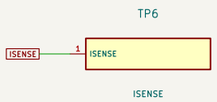


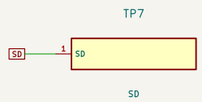


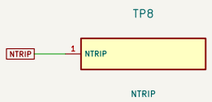


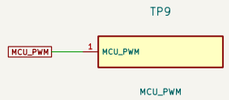










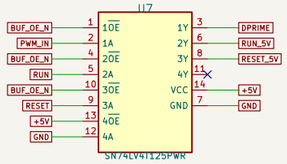


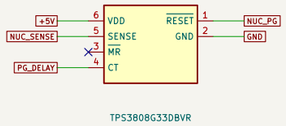
UNAL Manizales – Sistemas Dinamicos y Control

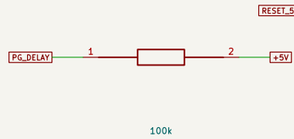
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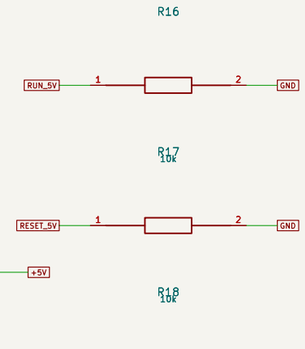
Title: STM32 interface and test points – REV B PROTOTYPE

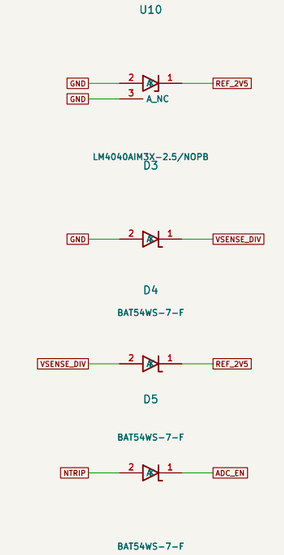
Size: A3	Date:	**Rev: B**
KiCad E.D.A. 10.0.7		Id: 5/6











Rev B power-domain interface. U7 buffers PWM/RUN/RESET from the NUCLEO 3.3 V domain into the Buck 5 V logic; outputs are high-impedance unless U9 sees NUC_3V3 >= 3.07 V (BUF_OE_N via spare U3 gate). R3 keeps DPRIME high (switch off), R16/R

UNAL Manizales – Sistemas Dinamicos y Control		
Sheet: /Power-domain interface/		
File: buck_5.kicad_sch		
Title: Power-domain interface – REV B PROTOTYPE		
Size: A3	Date:	**Rev: B**
KiCad E.D.A. 10.0.7		Id: 6/6